

CHAPTER 1 SYSTEM OVERVIEW

IR-7000 Series Industrial Robot User Manual

Educational Sample Document · Not a manufacturer publication.

The IR-7000 is a six-axis articulated industrial robot designed for material handling, machine tending, and light assembly. The system consists of the manipulator arm, the RC-7000 controller, teach pendant TP-70, and optional safety fencing.

Axes:

- Axis J1: Base rotation
- Axis J2: Shoulder
- Axis J3: Elbow
- Axis J4: Wrist roll
- Axis J5: Wrist pitch
- Axis J6: Wrist twist / tool flange

Nominal reach: 1400 mm for IR-7000-14.

Repeatability: ± 0.05 mm under rated conditions.

Controller input power: 200-230 VAC three-phase, 50/60 Hz.

CHAPTER 2 OPERATING MODES

The RC-7000 controller supports the following operating modes:

TEACH MODE

Used for jogging axes and recording taught points. Enabling device (deadman switch) on the teach pendant must be held in the middle position.

AUTO MODE

Runs selected programs from the controller. External safety circuit must be closed. Teach pendant enabling device is not required in Auto when the protective enclosure is secured.

REMOTE MODE

Starts and stops programs through digital I/O or fieldbus commands from a cell PLC. Remote mode inherits Auto safety requirements.

Mode changes are performed from the teach pendant Mode key or the controller cabinet mode selector. An invalid mode transition raises alarm A-210.

CHAPTER 3 STARTUP PROCEDURE

Recommended cold start procedure:

1. Verify emergency stop buttons are released on the controller and pendant.
2. Confirm main disconnect is ON and control power LED is green.
3. Wait for controller boot complete message on the pendant (approximately 45 seconds).
4. Select Teach mode.
5. Press MOTOR ON and hold the enabling device.
6. If motors do not energize, check alarm history for SRVO or power faults.
7. Perform a slow jog of each axis through a small motion to confirm servo response.

Do not bypass the enabling device. Do not operate with safety gates open unless the system is intentionally placed in Teach with reduced speed.

CHAPTER 4 TOOL AND USER FRAMES

Tool Center Point (TCP) calibration defines the offset from the flange to the tool tip. An inaccurate TCP causes path deviation and may trigger collision detection.

User frames define work object coordinate systems. Recalibrate the user frame when the fixture is relocated. Use the three-point method described in the Calibration Manual section of the Maintenance Manual.

Maximum recommended tool mass for IR-7000-14 including gripper: 14 kg.

This payload rating assumes the tool center of gravity is within 100 mm of the flange axis. Exceeding the rated payload may cause SRVO-021 overload alarms and mechanical wear.